Open-Source Procedural Radio Propagation Modelling for Sensor Networks in Smart Agriculture

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***Abstract*—Wireless sensing for smart agriculture, such as in greenhouses and vertical farms, must operate through crop canopies whose geometry and electromagnetic properties change with growth and water content. The same propagation conditions affect sensor connectivity, radio frequency (RF) energy delivery, and the feasibility of sensing the crop through changes in the wireless channel. This paper presents HOUSIONNA, a workflow connecting procedural scene generation in Houdini with GPU-accelerated radio ray tracing in Sionna RT. Batch execution supports systematic exploration of alternative crop and infrastructure configurations. The evaluation stage combines visual inspection of paths and radio maps with application metrics for sensor coverage and RF energy harvesting; backscatter analysis additionally requires a tag and reader model. The workflow provides a reproducible simulation foundation towards agricultural digital twins.**

***Index Terms*—Smart agriculture, environmental monitoring, Internet of Things, sensor networks, RF energy harvesting, flexible plant sensors, wireless sensing, digital twins, ray tracing.**

I. Introduction

Controlled Environment Agriculture (CEA) relies on sensing and control to manage crop production in greenhouses and vertical farms. Spatial microclimate differences motivate measurements at multiple locations [1], while plant-mounted sensors provide observations of the crop itself. Leaf movement measurements can support studies of abiotic stress [2], and flexible sensors can monitor plant conditions [3]. An RF-powered batteryless plant movement sensor combines inertial measurements with Bluetooth Low Energy (BLE) telemetry [4]. Such devices must accommodate plant movement and growth, favouring small nodes with limited energy storage. Their placement is driven by the measurement task, which may put them deep inside a canopy.

These applications impose different RF requirements. Active telemetry needs sufficient link margin, and batteryless operation also requires enough harvested energy per observation. Backscatter reduces the need for a tag to generate its own carrier [5], but requires both illumination and a return link to the reader. Distributed wireless power transfer (WPT) depends on spatial channels and coherent combining [6]. The propagation channel can also provide observations of the environment, as illustrated by radio tomography [7]; relating channel changes to crop conditions requires crop-specific models and independent reference measurements.

The cultivation environment affects all these links. Leaves, stems, growing trays, support structures, and greenhouse surfaces alter line-of-sight visibility and multipath. Growth changes the geometry, while water content changes the electromagnetic properties of plant tissue. Measurements in a tomato greenhouse show substantial differences from conventional propagation models [8]. Deployment analysis therefore needs site-specific channels and should account for changes in the crop, device positions, and infrastructure.

Procedural scene generation makes these comparisons practical. Parameterised models can vary plant spacing, canopy dimensions, leaf orientations, growth stage, and device placement without manually rebuilding each scene. Explicit parameters make variations repeatable and enable controlled sweeps in which selected properties change while the remaining assumptions stay fixed. The same scene family can then support communication and energy analyses, exposing placements that work for one crop configuration but fail as the canopy develops. This reduces scene-preparation effort and supports reproducible comparisons of sensor and transmitter layouts.

This paper presents HOUSIONNA, a workflow connecting procedural scene generation in Houdini with GPU-accelerated radio ray tracing in Sionna RT [9]. It integrates scene preparation and export, batch execution, run provenance, visual inspection, and application analysis; Sionna RT provides the electromagnetic solver [10], [11]. Paths, link metrics, and radio maps can be inspected in Houdini or a web viewer and used to evaluate sensor coverage and RF energy harvesting. A batteryless plant-sensor use case brings these stages together to examine connectivity, sustainable reporting intervals, and transmitter placement.

Figure 1 outlines the workflow. Sections II to IV describe procedural scene generation, ray tracing, and evaluation, respectively. Section V applies the workflow to the plant-sensor use case, and Section VI discusses limitations and the steps towards an agricultural digital twin.

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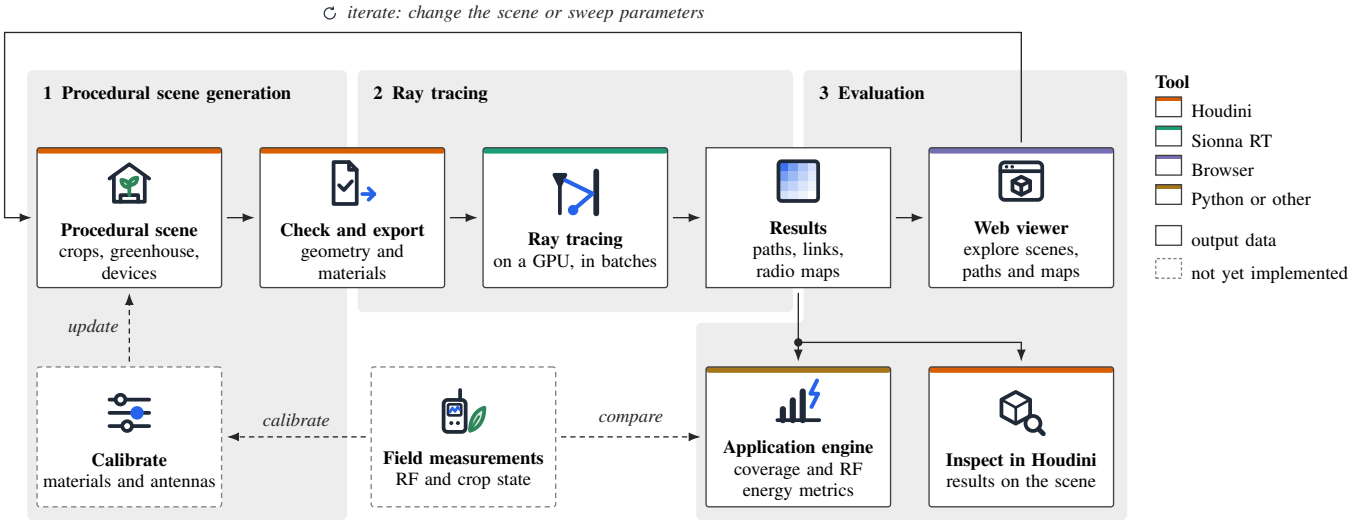


Fig. 1: The HOUSIONNA workflow in three stages. *Check and export* and *Results* hand over between stages; the bar on each step shows the tool that performs it, and the square box is output data. Dashed elements are needed for an agricultural digital twin, in which field measurements calibrate the materials and antennas of the scene. Each box and stage title links to the section that describes it.

II. Procedural Scene Generation

The first stage of HOUSIONNA constructs the agricultural environment in Houdini. A parameterised node network generates geometry, assigns material attributes, and places transmitters, receivers, and measurement surfaces for the ray-tracing stage. It can represent alternative crop layouts or infrastructure configurations without manually rebuilding every scene.

A. Agricultural Geometry

The scene parameters include plant spacing, canopy height and depth, leaf orientations, infrastructure layout, and device positions; transmitters and receivers can be placed anywhere, including on plants and on mobile robots. Plants are generated with L-systems in three growth stages, optionally with fruit, so growth is represented by a sequence of static scenes. People, autonomous robots, windows, and metal frames can be added as disturbances of the channel. A geometric growth parameter is not automatically a validated physiological model. **??** illustrates the three growth stages.

B. Radio Materials

The built-in ITU-R P.2040 materials of Sionna are bounded in frequency [12]: apart from wood and glass, they are defined only from 1 GHz, which excludes the 868 MHz and 915 MHz bands. HOUSIONNA therefore adds frequency-dependent models and records a source and status for each material: Cole–Cole fits of common building materials measured from 0.2 GHz to 67 GHz [13], conductors for metal structures, water and nutrient solution [14], soils [15], and the dual-dispersion vegetation model for leaves and stems [16]. The vegetation coefficients are not yet verified, and fruit and growing media use placeholder values, so plant materials must be treated as provisional. ITU-R P.833 provides an independent vegetation-attenuation reference for appropriate scenarios [17]; it does not replace calibration for the specific crop and geometry.

III. Ray Tracing

The second stage transfers the procedural scene to Sionna RT, computes propagation paths and radio maps, and records each run for subsequent inspection and analysis.

A. Scene Checks and Export

The exporter groups triangular geometry by object name and writes meshes, a scene description, material parameters, device definitions, and solver settings.

Export checks include triangulation, object names, material consistency, and device definitions. These are checks of simulation input integrity; they do not establish biological realism or material accuracy. A preview stage loads the exported scene in Sionna and returns its geometry for inspection in Houdini. This helps identify coordinate, scale, and export discrepancies before interpreting propagation results.

B. Solver Configuration and Geometry Interactions

The current implementation uses Sionna RT 2.2.0 and a single isotropic, vertically polarised antenna at each transmitter and receiver [9]. The solver accepts controls for line of sight, specular and diffuse reflection, transmission/refraction, diffraction, interaction depth, and sampling budgets. Enabling an interaction does not guarantee that a chosen scene representation models the corresponding canopy physics accurately.

Sionna models interactions with surfaces: each surface a ray crosses acts as a slab with the full thickness of its radio material. A closed canopy shell, internal sheets, and explicit leaf geometry can produce different depth-dependent attenuation. Refining the mesh alone does not resolve this modelling choice. When a leaf or stem is small relative to wavelength, a surface ray approximation may miss relevant

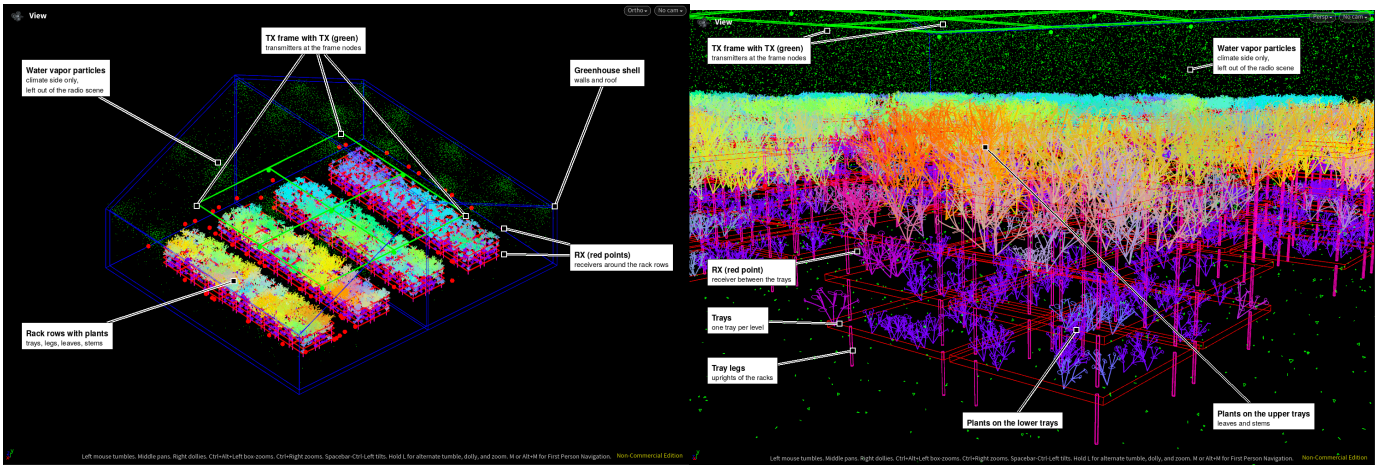


Fig. 2: Caption

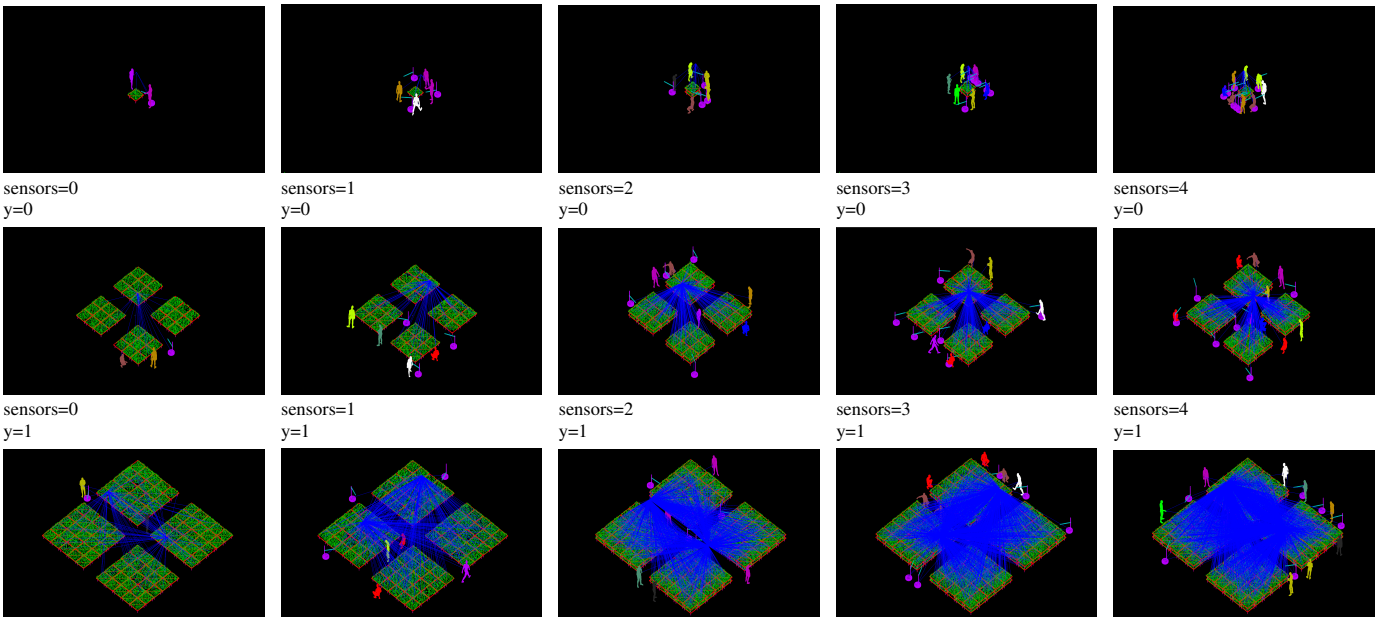


Fig. 3: Procedural scene variations.



Fig. 4: *Fragaria* × *ananassa*, variant 1.

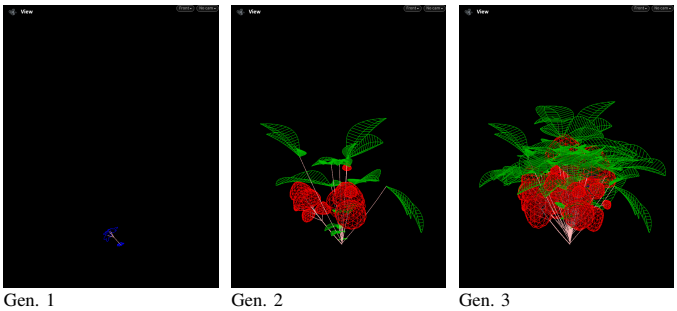
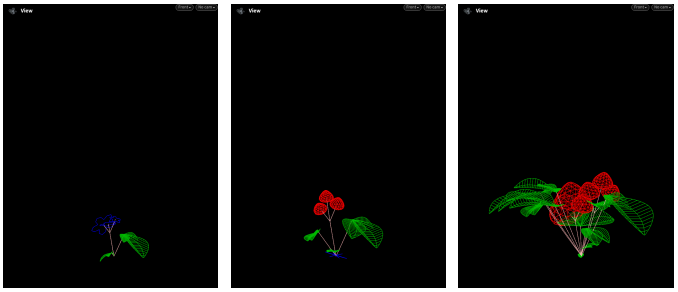


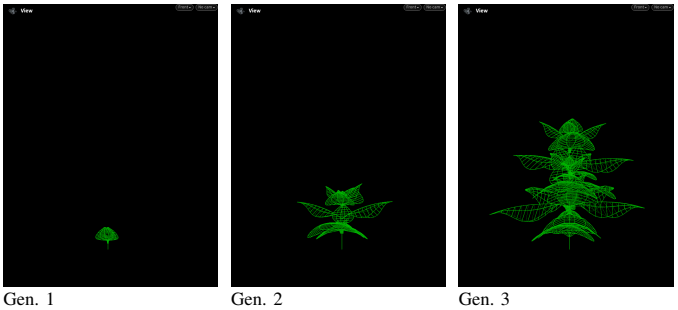
Fig. 5: *Fragaria* × *ananassa*, variant 2.

scattering behaviour. Sionna RT also discards paths containing a segment shorter than $1\,\mathrm{cm}$, so every path through a closed leaf or stem thinner than this is lost. Both effects were confirmed with analytical reference cases (free space,



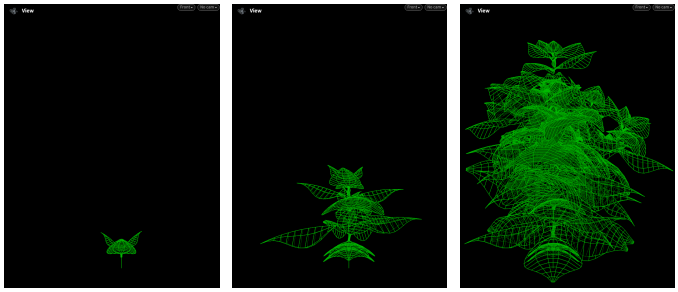
Gen. 1 Gen. 2 Gen. 3

Fig. 6: *Fragaria* × *ananassa*, variant 3.



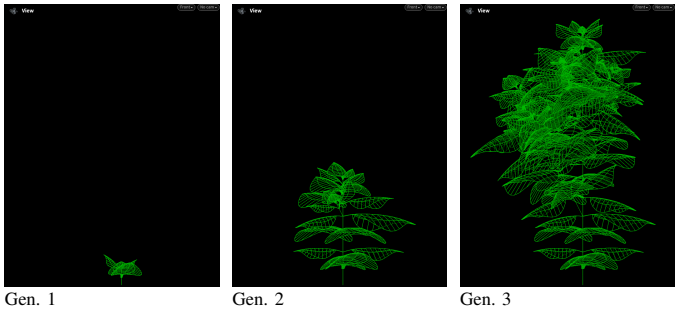
Gen. 1 Gen. 2 Gen. 3

Fig. 7: *Ocimum basilicum*, variant 1.



Gen. 1 Gen. 2 Gen. 3

Fig. 8: *Ocimum basilicum*, variant 2.



Gen. 1 Gen. 2 Gen. 3

Fig. 9: *Ocimum basilicum*, variant 3.

a single slab, a closed block, and thin glass) included in the repository [9]. Representation and material uncertainty must be assessed together.

C. Execution and Reproducibility

The solver runs in a separate Python environment, locally on an NVIDIA GPU or a CPU under Linux, or on a remote GPU host. Results are written as compressed NumPy arrays for the evaluation stage. This separation allows the modelling workstation and compute machine to differ.

For parameter sweeps, Houdini task-operator (TOP) work items, one per parameter combination, are exported to individual run folders. A background batch process solves work items sequentially within one Sionna session. Successful items can be skipped when their recorded solver settings match, and failed items are recorded for later retry.

Run records include the exported scene, resolved material properties, devices, settings, a solver copy, and software provenance. Batch records also preserve work-item parameters. Recording these assets makes the simulated geometry inspectable even after the procedural node network has changed. The records do not yet include the random seed of the path solver or hardware details, which a complete release should add.

IV. Evaluation

The third stage makes the simulated channels available for inspection and application analysis. Results can be examined in Houdini or in a web viewer, while an application engine converts the propagation outputs into relevant application metrics, e.g., coverage and energy metrics.

A. Returned Quantities

Propagation paths are imported as polylines carrying transmitter and receiver identifiers, complex amplitudes, delays, interaction types, and object associations. Per-link outputs include path counts, coherent and incoherent gains, RMS delay spread, and frequency-response magnitudes. Radio maps provide path gain per transmitter over triangular measurement surfaces.

B. Visual Inspection and Run Browsing

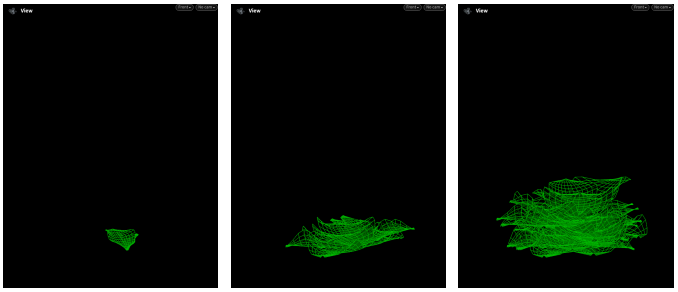
The compressed result arrays are imported into Houdini as attributed geometry, allowing paths and radio maps to be inspected in the original scene. A web viewer built on three.js displays the same paths, radio maps, and link metrics without a Houdini installation, steps through the runs of a batch, and can be published as a read-only website. This allows results to be disseminated more conveniently without relying on Houdini.

C. Application Engine and Metrics

The application engine converts the raw RF propagation transmitter–receiver channels into metrics relevant for different applications and technologies. For instance, it can evaluate the energy harvesting performance of the system as the plants grow.

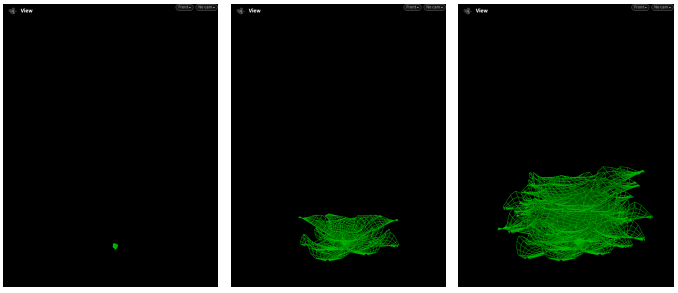
1) Active Radio Coverage: Let $g_{ij} = \sum_k |a_k|^2$ be the incoherent power gain over the paths k from transmitter i to receiver j, including the antenna assumptions of the simulation. With transmit power $P_{\mathrm{t},i}$ in dBm, the link margin in decibels is

$$M_{ij} = P_{\mathrm{t},i} + 10\log_{10} g_{ij} - S_j - L_{\mathrm{impl}} - M_{\mathrm{reserve}}, \quad (1)$$



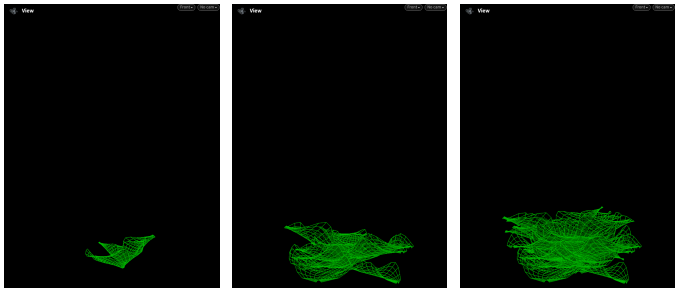
Gen. 1 Gen. 2 Gen. 3

Fig. 10: *Lactuca sativa*, variant 1.



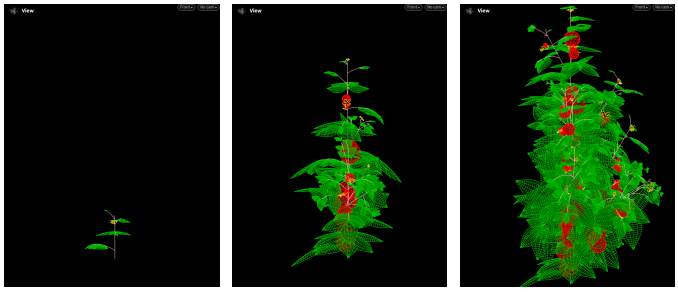
Gen. 1 Gen. 2 Gen. 3

Fig. 11: *Lactuca sativa*, variant 2.



Gen. 1 Gen. 2 Gen. 3

Fig. 12: *Lactuca sativa*, variant 3.



Gen. 1 Gen. 2 Gen. 3

Fig. 13: *Solanum lycopersicum*, variant 1.

where S_j is the receiver sensitivity, L_{impl} covers implementation losses not already included in g_{ij}, and M_{reserve} is a design reserve. Because the channel is reciprocal, the same gain gives the uplink margin of a node, with P_{t} the node's transmit power.

For a scenario set $\mathcal{S}$, a useful placement criterion is

$$M_{\text{worst}} = \min_{s\in\mathcal{S}} \min_{j} \max_{i} M_{ij}^{(s)}. \tag{2}$$

This tests whether each required sensing location has at least one adequate infrastructure link across the considered crop configurations. It does not predict interference, network capacity, or protocol reliability.

BLE and LoRa link budgets require their own transmit powers and receiver sensitivities. Flexible or plant-mounted devices likewise require antenna, mounting, and energy parameters consistent with the hardware being evaluated.

2) RF Power and Transmitter Strategies: For independent or time-separated sources, received powers may be combined under the corresponding averaging assumptions. Coherent multi-source operation instead requires

$$P_{\text{RF},j} = \left|\sum_i h_{ij} w_i\right|^2, \qquad \sum_i |w_i|^2 = P_{\text{TX}}, \tag{3}$$

where $h_{ij} = \sum_k a_k e^{-j2\pi f \tau_k}$ is the narrowband channel, summed coherently over the paths with amplitudes a_k and delays τ_k,[1] and w_i is a transmit weight with units $\sqrt{\text{W}}$. With N transmitters and the same total power P_{TX}, the engine compares four strategies: all power on the best transmitter, $P_{\text{TX}} \max_i |h_{ij}|^2$; an equal split without phase alignment, $(P_{\text{TX}}/N)\sum_i g_{ij}$; an equal split in phase without channel knowledge, $(P_{\text{TX}}/N)|\sum_i h_{ij}|^2$; and maximum-ratio transmission (MRT) with $w_i \propto h_{ij}^*$, which gives $P_{\text{TX}}\sum_i |h_{ij}|^2$. MRT assumes ideal channel knowledge at all transmitters and focuses the power on one receiver at a time, so it is an upper bound for distributed operation [6]. Each transmitter has a single isotropic antenna; arrays per transmitter and the validation of simulated phases remain future work.

3) Energy Harvesting and Reporting Interval: For WPT, propagation gain must be combined with a measured rectifier or harvester response. If F_{EH} maps incident RF power and frequency to usable DC power, an energy balance over an observation period T is

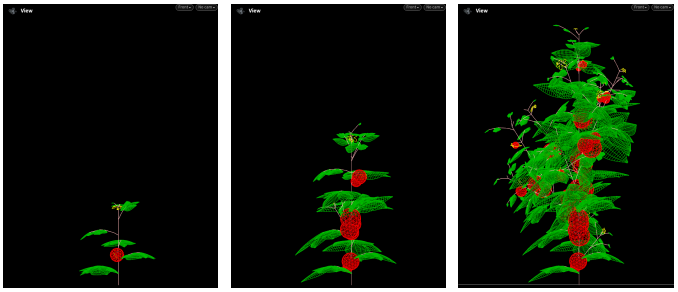
$$\int_0^T F_{\text{EH}}(P_{\text{RF}}(t), f)\, dt \geq E_{\text{cycle}} + E_{\text{loss}}. \tag{4}$$

The response should account for startup, storage state, and operating conditions when these affect the device. A constant conversion efficiency is generally insufficient near the operating threshold.

For a static channel, the shortest sustainable reporting interval is $T_{\min} = E_{\text{cycle}}/F_{\text{EH}}(P_{\text{RF}}, f)$ when F_{EH} represents the net power stored in the buffer, including conversion losses, leakage, and quiescent consumption. Device-specific responses therefore determine how RF power maps to sensing performance.

Reference sensor and rectifier models: The plant sensor from [4] provides a concrete energy-budget example: the reported measurement-and-transmission cycles require $2.88\,\text{mJ}$ or $9.11\,\text{mJ}$, depending on the selected measurements. We de-

[1]The path coefficients of Sionna RT exclude the propagation phase, so the factor $e^{-j2\pi f\tau_k}$ must be applied explicitly.



Gen. 1 Gen. 2 Gen. 3

Fig. 14: *Solanum lycopersicum*, variant 2.



Gen. 1 Gen. 2 Gen. 3

Fig. 15: *Solanum lycopersicum*, variant 3.

rive F_{EH} from the measured buffer charge times of this sensor at 915 MHz [4], which include conversion losses, leakage, and quiescent currents: nothing is harvested below $-15\,\mathrm{dBm}$, and the net efficiency rises from 6 % at that threshold to 30 % at $0\,\mathrm{dBm}$.

As an upper bound, we use measurements of the rectifier alone, the high-power input path of the 915 MHz AEM30940 evaluation kit used by the sensor, at its best load voltage [18]. It converts 11 % at $-15\,\mathrm{dBm}$ but less than 2 % at $-17.5\,\mathrm{dBm}$, so the threshold of the sensor is set mainly by the rectifier, while the charger roughly halves the efficiency above it. The low-power path of the same kit, intended for inputs below $-10\,\mathrm{dBm}$, still converts 23 % at $-20\,\mathrm{dBm}$ and is evaluated as a design alternative.

4) Transmitter Placement: Candidate transmitter subsets are compared using the precomputed channels, applying the same communication and energy requirements to each subset. Reporting the preferred subset for each number of transmitters shows whether the locations that support connectivity also provide sufficient RF power. The comparison uses the coverage criterion in (2) and the reporting intervals derived from (4). Candidate positions, the number of active transmitters, and the required margins or reporting intervals are specified for each deployment.

5) Backscatter Extension: Illumination-to-tag and tag-to-reader gains can be obtained from one run by reciprocity. A tag and reader model must be added to convert these channels into a modulated received signal; the current engine does not implement a modulating tag.

A simple narrowband bistatic model is

$$P_{\mathrm{mod},r} \approx P_{\mathrm{ill}}\, g_{\mathrm{ill},t}\, g_{t,r}\, \eta_{\mathrm{mod}}, \tag{5}$$

where the two gains describe illumination-to-tag and tag-to-reader propagation, and η_{mod} represents effective differential tag modulation under consistent antenna normalisation. A complete model needs tag-specific scattering, polarisation, reader noise, and the direct illuminating signal.

Both gains follow from a single run in which the illuminator and the reader are transmitters and the tag locations are receivers, since the channel is reciprocal. The engine does not yet implement a modulating tag; a backscatter study therefore needs a separate tag and reader model and validation against the deployed geometry.

V. Use Case

The use case considers a deployment of batteryless plant sensors in CEA. The sensor and harvester models, evaluation metrics, and transmitter strategies are described in Section IV-C; this section specifies the agricultural scene, device placement, and simulation settings, and reports the resulting coverage and reporting intervals.

A. Crop and Cultivation Environment

gilles Specify the plant species, growth stages, canopy dimensions, and plant spacing. Describe the greenhouse or vertical-farm geometry, cultivation infrastructure, and material assumptions used in the selected runs.

B. Sensor and Transmitter Locations

gilles Specify the number of sensors and their positions on the plants, including canopy height, depth, and mounting orientation. If environmental sensors are included, give their microclimate sampling locations. List candidate gateway and power-transmitter positions and the transmitter counts considered.

C. Simulation and Device Settings

gilles State the simulated frequencies, transmit powers, receiver sensitivities, antenna assumptions, solver settings, and batch variations. Identify the sensor and harvester response selected from Section IV-C; the measured plant-sensor model is specific to 915 MHz. Give the target link margin and reporting interval, and the transmitter-subset search settings used for this deployment.

D. Deployment Results

gilles Insert coverage maps and worst-case uplink margins, identifying failures by sensor location and crop configuration. Add reporting-interval maps and the area share meeting the target interval for each transmitter strategy and selected harvester model. Report the selected transmitter positions and performance versus transmitter count. Numerical results are pending.

VI. Discussion and Conclusions

HOUSIONNA connects procedural scene generation, GPU radio ray tracing, batch execution, inspection of paths and radio maps in Houdini and a web viewer, and an application engine for communication and WPT metrics. This provides a practical basis for studying the RF infrastructure serving agricultural sensors across changing crop configurations. Environmental monitoring, plant health monitoring, batteryless IoT, and remote canopy sensing share this propagation foundation while requiring different application models. Two findings hold independently of the scene: surface-based ray tracing loses every path through structures thinner than $1\,\mathrm{cm}$, and for the measured batteryless plant sensor the rectifier, rather than the channel alone, sets the input power at which the node can operate.

A. Towards an Agricultural Digital Twin

Procedural simulation supports controlled comparisons and reproducible scenario generation, but a digital twin also requires a maintained relationship with the physical environment. In this setting, that relationship would connect observed crop geometry and operating conditions to calibrated RF materials, antenna models, and validated predictions. Periodic updates could then test whether an existing network remains adequate as the crop develops.

Three limits are particularly relevant. First, the present vegetation materials are provisional, and surface-based canopy representations need evaluation across the intended frequencies, in particular for thin stems and leaves. Second, the isotropic single-antenna configuration does not capture plant-mounted antenna detuning, realistic polarisation effects, or array-based power delivery. Third, a sequence of static scenes does not provide continuous state estimation or validated dynamic sensing.

Sionna RT supports differentiable modelling [10], [11], but the file-based wrapper described here does not provide an end-to-end differentiable Houdini pipeline. The use case compares transmitter placement by searching subsets of candidate positions on precomputed channels; gradient-based material fitting or continuous placement would need a deliberate integration with the solver. Protocol performance, backscatter modulation, and RF crop inference likewise remain separate modelling tasks.

The broader HyROPA programme includes optical propagation. Here the RF model is the focus; optical energy harvesting and hybrid links are possible later extensions that should reuse compatible geometry and observations without assuming identical interaction models.

B. Future Work

The next scientific step is to validate canopy representations and material parameters, then evaluate communication, energy, and sensing metrics against physical observations. This progression supports the paper's direction towards agricultural digital twins without treating procedural visualisation alone as a validated twin.

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